

① $F = \mathbb{C} \Rightarrow G = \langle 1 \rangle$.

$F = \mathbb{R} \quad f(x) = 3x^2 + 2 > 0 \Rightarrow f(x)$ has 1 real root $\Rightarrow G = \mathbb{Z}/2\mathbb{Z}$

$F = \mathbb{Q} \quad \Delta = -4 \cdot 8 - 27 \cdot 4 = -4 \cdot 35$ not a square in $\mathbb{Q}$ so $G = S_3$

$F = \mathbb{Z}/3\mathbb{Z} \quad f(0)=1 \quad f(1)=1 \quad f(2)=1 \Rightarrow f(x)$ has no zeroes in F

$\Rightarrow f(x)$ is irreducible $\Rightarrow G = \mathbb{Z}/3\mathbb{Z}$

② $f(x) = x^5 - 2$ is irred by Eisenstein with $p=2$. Let α be a root, $\alpha = \sqrt[5]{2}$

then $K = \mathbb{Q}[\sqrt[5]{2}, \zeta_5]$ where ζ_5 is a prim. 5th root of 1.

Now $[\mathbb{Q}(\sqrt[5]{2}), \mathbb{Q}] = 5$ $[\mathbb{Q}(\zeta_5), \mathbb{Q}] = 4$ are rel. prime & so $[K, \mathbb{Q}] = 20$. $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ is Galois so $\mathbb{Q}(\zeta_5) = K^H$ for $H \triangleleft G(K, \mathbb{Q})$

$G(\mathbb{Q}(\zeta_5): \mathbb{Q}) \cong \mathbb{Z}/4\mathbb{Z} \cong G(K, \mathbb{Q})/H \Rightarrow |H| = 5$ i.e. $H \cong \mathbb{Z}/5\mathbb{Z}$

G is not abelian as $\mathbb{Q} \subset \mathbb{Q}(\sqrt[5]{2}) \subset K$ and $\mathbb{Q}(\sqrt[5]{2})$ is not normal.

③ Let $d = \deg f(x)$. $F[x]/f(x) \cong F_q^d$ the finite field with q^d elements

If d divides n , then $F_q^d \subset F_q^n \subset F_q^a$ & $F_q^a = \left\{ \begin{array}{l} \text{solutions to } x^{q^n} - x = 0 \\ \text{in } F_q^a \end{array} \right\}$

so $f(x) = \text{luc}(\alpha, x, F_q)$ some $\alpha \in F_q^d \subset F_q^n \Rightarrow f(x) \mid x^{q^n} - x$

If $f(x) \mid x^{q^n} - x$ then $F[x]/f(x)$ is a subfield of F_q^n containing F_q

Let $q = p^t$ $p = \text{char } F$. Then we have seen in class that $F[x]/f(x) \cong F_{p^s}$ where $t \mid s$ & $s \mid tn$ so $s = td$ where $d \mid n$

i.e. $F[x]/f(x) \cong F_{p^d}$ where $d \mid n$.

④ $G(\mathbb{Q}(\zeta_m): \mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^*$ if $m = 2g$ then m is prime so

$\phi(2g) \cong 28$ so $G(\mathbb{Q}(\zeta_{2g}): \mathbb{Q}) \cong \mathbb{Z}/28\mathbb{Z} = G$ pick $H = \mathbb{Z}/7\mathbb{Z} \subset \mathbb{Z}/28\mathbb{Z}$

then $\mathbb{Q}(\zeta_{2g})^H = K$ is Galois over $\mathbb{Q}$ & $G(K, \mathbb{Q}) = \mathbb{Z}/28\mathbb{Z} / \mathbb{Z}/7\mathbb{Z}$

$= \mathbb{Z}/7\mathbb{Z}$ as required.